

MAT1100 Homework 2

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Question 1

- (a) *Proof.* Suppose $K, N \triangleleft G$ with $K \leq N$.

First we show that

$$N/K \triangleleft G/K$$

suppose $g \in G$ and $n \in N$ then

$$[g]_K [n]_K [g^{-1}]_K = [gn g^{-1}]_K \in N/K$$

where final part follows from the fact that $N \triangleleft G$ so $gn g^{-1} \in N$.

Next, to construct a isomorphism from

$$(G/K)/(N/K) \cong G/N$$

it suffices to find a surjection

$$\varphi : G/K \rightarrow G/N$$

whose kernel is N/K and factor this map through the quotient by its kernel.

Concretely, define

$$\varphi : [g]_K \rightarrow [g]_N$$

first to see that φ is well defined: suppose $g, h \in G$ such that $[g]_K = [h]_K$ which means that $g^{-1}h \in K \leq N$. Therefore

$$[g]_N = [h]_N$$

hence φ is well defined.

That φ is surjective is clear, as if $g \in G$

$$\varphi([g]_K) = [g]_N$$

by definition.

Next, we show that $\ker \varphi = N/K$. It is clear that $N/K \subseteq \ker \varphi$. As if $n \in N$, $[n]_N = e$ and $\varphi([n]_K) = e$. Next if $[g]_K \in \ker \varphi$ then

$$[g]_N = e$$

and hence $g \in N$ hence $[g]_K \in N/K$ as required.

Now by the first isomorphism theorem

$$(G/K)/(N/K) \cong (G/N)$$

with an explicit isomorphism given by

$$\hat{\varphi} : [[g]_K]_{N/K} \mapsto \varphi([g]_K) = [g]_N$$

□

- (b) *Proof.* Suppose $K = \{e, (12)(34), (13)(24), (14)(23)\} \leq S_4$. This is a normal subgroup as conjugation preserves cycle type and K is the union of the conjugacy classes (1) and (2, 2).

Now $|S_4| = 24$ and $|K| = 4$. By Lagrange's theorem, if H is a normal subgroup of S_4 containing K its order must be divisible by 4 and divide 24. Hence $|H| \in \{4, 8, 12, 24\}$. Now since H is normal it is the union of conjugacy classes in S_4 . the conjugacy classes of S_4 correspond to the cycle types (1), (2), (2, 2), (3), (4) with sizes 1, 6, 3, 8, 6. K is already the union of (1) and (2, 2) so by inspection it is clear that either $|H| = 4$ in which case $H = K$, or $|H| = 12$ in which case $H = A_4$ or $|H| = 24$ in which case $H = S_4$.

If $H = K$ then $H/K \cong \{e\}$.

Else, if $H = A_4$ then $|A_4/K| = 3$ hence by Lagrange's theorem

$$A_4/K \cong \mathbb{Z}/3\mathbb{Z}$$

Finally if $H = S_4$ then we can use the action defined in lecture by $S_4 \curvearrowright \Delta$ where

$$\Delta = \{P_1, P_2, P_3\}$$

with

$$P_1 = \{\{1, 2\}, \{3, 4\}\}$$

$$P_2 = \{\{1, 3\}, \{2, 4\}\}$$

$$P_3 = \{\{1, 4\}, \{2, 3\}\}$$

and S_4 permutes the individual coordinates of each P_i . In lecture we showed that if $\pi : S_4 \rightarrow S_\Delta \cong S_3$ is the permutation representation of this action, $\ker \pi = K$ and so $|S_4/K| = 6$ and since $|S_3| = 6$ this implies that π is surjective and hence

$$S_4/K \cong S_3$$

Now to construct an explicit isomorphism

$$(S_4/K)/(A_4/K) \cong \{-1, 1\}$$

we use question part (a) and map

$$\varphi : (S_4/K)/(A_4/K) \rightarrow S_4/A_4$$

given by

$$\varphi([\sigma]_K)_{A_4/K} = [\sigma]_{A_4}$$

but since

$$S_4/A_4 \cong \{-1, 1\}$$

via the sign map we can define an isomorphism

$$\psi : (S_4/K)/(A_4/K) \rightarrow \{-1, 1\}$$

by

$$\psi([\sigma]_K)_{A_4/K} = \text{sgn}(\sigma)$$

□

- (c) *Proof.* Take $L = \{e, (12)(34)\} \leq K$ which is normal in K but not normal in G as conjugating by (23) we get (13)(24) which is not in L . This means that G/L is not a group and so the statement

$$(G/L)/(K/L) \cong G/K$$

is not well defined as the set being quotiented is not a group.

□